

DESHAN SIRIWARDHANA

Full-Stack Developer | Undergraduate

BSc (Hons) Software Engineering | CINEC Campus

CONTACT

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LOCATION

Kegalle / Colombo, Sri Lanka

PORTFOLIO

deshansiriwardhana.me

GITHUB

github.com/deshan2004

LINKEDIN

<http://www.linkedin.com/in/deshan-siriwardhana>

EDUCATION

CINEC Campus

2024 - Present | Malabe, Sri Lanka

BSc (Hons) in Software Engineering

Key coursework: Data Structures & Algorithms, Software Construction and Evolution, Software Quality Assurance, Database Management Systems, Mobile Computing, Advanced Web Technologies, Software Testing.

Swarna Jayanthi College

2022 | Kegalle, Sri Lanka

G.C.E. Advanced Level - Physical Science Stream

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Java, Python, C#, PHP, C, Kotlin, Dart

Frontend & Mobile: Next.js, React.js, Angular, Vue.js, Flutter, Tailwind CSS, Bootstrap, HTML5, CSS3

Backend & APIs: Node.js, Express.js, RESTful APIs, WebRTC, ASP.NET Core, PHP, Nginx, Python, Java

Databases & Cloud: PostgreSQL, Supabase, MySQL, MSSQL, MongoDB, Firebase, Oracle

Testing & QA: Unit Testing, JUnit 5, Apache JMeter, Selenium WebDriver, Requirements Traceability (RTM)

DevOps & Tools: Git, GitHub, VS Code, Postman, IntelliJ IDEA, Android Studio, Figma, Vercel

OTHER SKILLS

Leadership & Management: Team Leadership, Agile / Scrum Practices, Project Planning, Cross-Functional Collaboration

Problem Solving: Analytical Thinking, Debugging, Critical Decision Making, Requirements Analysis

Communication & Interpersonal: Technical Documentation, Stakeholder Communication, Public Speaking, Client Engagement

Workplace Competencies: Time Management, Adaptability, Active Listening, Conflict Resolution

PROFILE

Results-oriented Software Engineering undergraduate and Full-Stack Developer specializing in modern JavaScript/TypeScript ecosystems, component-driven frontend frameworks (Next.js, React.js, Angular), and robust API backends (Node.js, Express, .NET Core). Experienced in architecting end-to-end web and mobile applications incorporating real-time communications, AI services, and relational/NoSQL databases (PostgreSQL, Supabase, Firebase, MongoDB). Committed to engineering clean, type-safe, maintainable software within Agile environments supported by rigorous unit testing and automated QA practices.

EXPERIENCE

Co-Founder & Technical Lead

2026 - Present

CodeCraft Solutions

- Collaborated on building customized responsive web applications and full-stack software solutions for local business requirements.

FEATURED PROJECTS

ShelfLife AI - AI-Powered Inventory & Expiry Loss Prevention Platform

React.js(vite) · Node.js · Firebase · Tesseract.js · REST API · PWA · JavaScript

Engineered an AI-OCR scanning module using Tesseract.js/Vision AI to auto-extract expiration dates directly from packaging, cutting manual stock entry time by 90%.

Developed a real-time FEFO (First-Expired, First-Out) inventory engine with tiered alerts (30/7/2-day windows) and batch-level supplier tracking to eliminate expiry losses.

Architected automated Flash Sale triggers that dynamically apply promotional discounts (BOGO, 30-50% off) on near-expiry items to recover at-risk revenue.

Integrated an AI conversational assistant & recipe engine to recommend bundled promotions and waste-reduction strategies for food/retail inventory.

Built an Offline-First PWA architecture with background data sync and role-based access control (User & Admin portals) for uninterrupted shop-floor operations.

Kingswood Connect - Real-Time Student Attendance & Management Platform

React.js(Vite) · Node.js · Express.js · Firebase · Socket.IO · Tailwind CSS · JavaScript

Engineered a hardware-free, full-stack student attendance and administrative management platform for educational institutions.

Architected a real-time mobile scanning integration using Socket.IO WebSockets, allowing smartphone cameras to function as live ID/QR scanners for desktop dashboards.

Configured a Node.js and Express backend integrated with Firebase (Firestore & Auth) to enforce multi-role authorization (Admin, Teacher, Student) and secure data management.

Designed an administrative web portal for live attendance analytics, automated student fee tracking, class scheduling, and a soft-delete data recovery pipeline using React and Tailwind CSS.

CERTIFICATIONS

Basics of Machine Learning Algorithms (Level 7)

Athena Global Education

Decision Trees, Naive Bayes, Predictive Modeling

Operate with DevOps

Microsoft

DevOps Practices, Continuous Integration & Delivery, Agile Workflows

Explore Microsoft 365 Security Foundations

Microsoft

Identity & Access Management, Threat Protection, Cloud Security

CIRS 2026 Workshop on Authentic Learning

CINEC Campus

Real-World Experiences That Build 21st-Century Skills

LANGUAGES

English: Professional Working Proficiency

Sinhala: Native / Bilingual Proficiency

REFERENCES

Mr. D.P.N. Hasaranga

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Neth-Sawan - AI-Powered Assistive Web Platform

React.js(vite) · JavaScript · Node.js · WebRTC · Google Generative AI SDK · Firebase

Engineered a real-time assistive communication portal tailored for hearing-impaired users to bridge daily communication barriers.

Implemented peer-to-peer real-time video communication using WebRTC paired with Firebase signaling servers.

Integrated Google Generative AI SDK to process camera feeds into instant visual contextual descriptions, ambient event detections, and scannable visual alerts.

Developed a high-contrast, accessible UI in React.js and TypeScript following WCAG accessibility guidelines.

KethPiyasa - Direct Farmer-to-Buyer B2B Agricultural Marketplace

Next.js · TypeScript · Node.js · Express.js · Firebase · Tailwind CSS

Engineered a full-stack B2B digital marketplace designed to streamline Sri Lanka's agricultural supply chain by bridging smallholder farmers directly with commercial buyers.

Implemented role-based portals for Farmers, Buyers, Logistics, and Admins featuring real-time crop bidding, counter-offer negotiations, and interactive geographic discovery maps.

Integrated an escrow holding payment workflow paired with automated invoice generation and QR code delivery verification to secure transactions.

Architected a trilingual localization system (English, Sinhala, Tamil) and live economic center baseline price tracking to ensure market transparency and accessibility.

RIGA Academy - Dance Academy Management & Online Enrollment Platform

Next.js · TypeScript · Firebase (Auth & Firestore) · Tailwind CSS · Framer Motion

Engineered a full-stack dance academy platform to streamline student enrollments, class scheduling, event showcases, and studio space management.

Integrated Firebase Authentication and Firestore DB to enforce multi-role access control (Admin & Student), securing user data and administrative permissions.

Developed an online enrollment & verification workflow, allowing students to register for dance packages, upload payment slips, and track registration status via a dedicated dashboard.

Architected an intuitive Admin Portal for live attendance tracking, class scheduling, payment slip approvals, user role management, and inquiry handling.

Crafted a modern, responsive UI/UX using Next.js, TypeScript, Framer Motion, and Tailwind CSS with dynamic animations, interactive schedules, and media galleries.

LEADERSHIP & VOLUNTEERING

Active Member

2026 - Present

AIESEC in CINEC

Contributing as an active member supporting youth leadership initiatives and chapter activities.